

Chapter 8

Partial Derivatives

We already know if given $f(x)$, we can find $f'(x)$

If the function have two variables, $f(x, y)$, we can treat one variable as a constant, then we can get partial derivatives with respect to x or y

$$\frac{\partial f}{\partial x} \quad \frac{\partial f}{\partial y}$$

$$dx = \Delta x \quad dy = \Delta y$$

$$\frac{dy}{dx} = \lim_{\Delta x \rightarrow 0} \frac{\Delta y}{\Delta x}$$

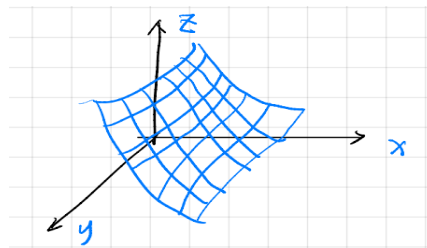
Example

$$f(x) = \sin x$$

$$f'(x) = \cos x = \frac{df}{dx}$$

$$df = \cos x \cdot dx \quad df = \frac{df}{dx} \cdot dx$$

For a two-dimension plain, $z(x, y)$

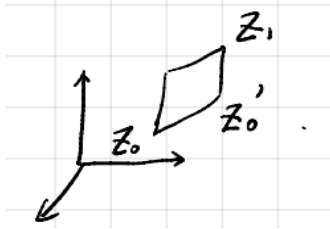


$$\frac{\partial z}{\partial x} \quad \frac{\partial z}{\partial y}$$

We have,

$$dz = \frac{\partial z}{\partial x} dx + \frac{\partial z}{\partial y} \cdot dy$$

Proof



$$\begin{aligned}
 z_0 &= z(x_0, y_0) & z_0' &= z(x_0 + \Delta x, y_0) \\
 z_1 &= z(x_1, y_1) \\
 dz &= z_1 - z_0 \\
 &= (z_1 - z_0') + (z_0' - z_0) \\
 &= \frac{\partial z}{\partial y} \cdot \Delta y + \frac{\partial z}{\partial x} \cdot \Delta x
 \end{aligned}$$

$$dz = \frac{\partial z}{\partial x} \cdot dx + \frac{\partial z}{\partial y} \cdot dy$$

□

1. Total Differentials 全微分

$$dz = \frac{\partial z}{\partial x_1} \cdot dx_1 + \frac{\partial z}{\partial x_2} dx_2 + \cdots + \frac{\partial z}{\partial x_N} dx_N$$

Example find $\frac{1}{\sqrt{0.25-10^{-20}}} - \frac{1}{\sqrt{0.25}}$

$$y = f(x), \quad \Delta y = f'(x) \Delta x$$

$$f(x) = \frac{1}{x^{0.5}}, \quad x_0 = 0.25, \quad \Delta x = 10^{-20}$$

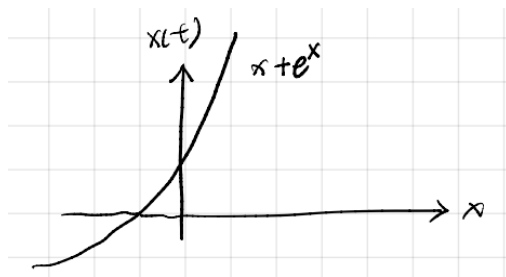
$$f(x_0 + \Delta x) - f(x_0) = f'(x_0) \cdot \Delta x = \left(-\frac{1}{2}\right) \cdot x^{-\frac{3}{2}}|_{x=0.25} \cdot (-10^{-20}) = 4 \times 10^{-20}$$

Example Demonstrate $\frac{1}{n^2} - \frac{1}{(n+1)^2} = \frac{2}{n^3}$

$$f(x) = \frac{1}{x^2}$$

$$-[f(n+1) - f(n)] = f'(x)|_{x=n} \cdot 1 \times (-1) = (-2) \cdot \frac{1}{n^3} \cdot (-1) = \frac{2}{n^3}$$

Example



$$\begin{aligned}
 x + e^x &= t \\
 x(t) &\rightarrow \frac{dx}{dt} \\
 \frac{dx}{dt} + e^x \frac{dx}{dt} &= 1 \\
 \frac{dx}{dt} &= \frac{1}{1 + e^x}
 \end{aligned}$$

Also,

$$\begin{aligned}
 \frac{dx^2}{dt^2} + e^x \frac{dx}{dt} \frac{dx}{dt} + e^x \frac{d^2x}{dt^2} &= 0 \\
 \frac{d^2x}{dt^2} &= \frac{-e^x}{(1 + e^x)^3}
 \end{aligned}$$

Example $x^3 - 3y^3 + xy + 21 = 0$ find tangent line

$$\begin{aligned}
 \frac{d}{dx} (x^3 - 3y^3 + xy + 21) &= 3x^2 - 9y^2 \frac{dy}{dx} + y + x \frac{dy}{dx} = 0 \\
 \frac{dy}{dx} &= \frac{-y - 3x^2}{x - 9y^2}
 \end{aligned}$$

If $(x, y) = (1, 2)$

$$\begin{aligned}
 \frac{dy}{dx} &= \frac{1}{7} \\
 \frac{y - 2}{x - 1} &= \frac{1}{7} \\
 \Rightarrow y &= x - 4
 \end{aligned}$$

Example $z = xy, x = \sin(s + t), y = s - t$, find $\frac{\partial z}{\partial s}$ and $\frac{\partial z}{\partial t}$

$$\begin{aligned}
 dz &= \frac{\partial z}{\partial x} dx + \frac{\partial z}{\partial y} dy & dz &= ydx + xdy \\
 dx &= \cos(s + t)ds + \cos(s + t)dt \\
 dy &= ds - dt
 \end{aligned}$$

$$\begin{aligned}\Rightarrow dz &= [y\cos(s+t) + x]ds + [y\cos(s+t) - x]dt \\ &= [(s-t)\cos(s+t) - \sin(s+t)]ds + [(s-t)\cos(s+t) - \sin(s+t)]dt\end{aligned}$$

Example $z = x - y, x^2 + y^2 = t^2, xsint - ye^y$, find $\frac{dz}{dt}$

First find total derivative

$$\begin{aligned}dz &= dx - dy \\ 2xdx + 2ydy &= 2tdt \\ sintdx + xcostdt &= (e^y + ye^y) \cdot dy\end{aligned}$$

Then get rid of dx, dy . According to the second and the third equation, we can get

$$\begin{cases} xdx + ydy = tdt \\ sindx - (y+1)e^y dy = xcostdt \end{cases}$$

$$dx = \frac{-t(y+1)e^y + xycost}{-x(y+1)e^y - ysint} dt$$

We can also get dy and dz

Then we can find $\frac{dz}{dt}$

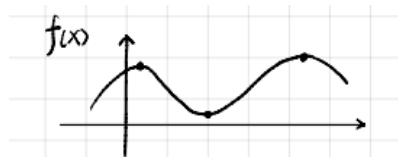
Example $z = x^2 + xy, x^2 + y^2 = st + 5, x^2y^2 = s^2 + t^2$ find $\frac{\partial z}{\partial t}$ and $\frac{\partial z}{\partial s}$

$$\begin{aligned}dz &= (2x + y)dx + xdy \\ 2xdx + 3y^2dy &= tds + sdt \\ 3x^2dx - 2ydy &= 2sds + 2tdt\end{aligned}$$

Example $w = x + y, x^2 + xy + y^2 = s, x^2y + xy^2 = t$, find $\frac{\partial w}{\partial s}$ and $\frac{\partial w}{\partial t}$

$$\begin{aligned}dw &= dx + dy \\ (3x^2 + y)dx + (x + 2y^2)dy &= ds \\ (2xy + y^2)dx + (x^2 + 2xy)dy &= dt\end{aligned}$$

2. Lagrange Multiplier



If we want to find ma/minimum point , we can try optimization.

If we can find the max of $f(x)$, then we can find its minimum, because

$$\max f(x) \Leftrightarrow \min -f(x)$$

We can find its min by critical point condition

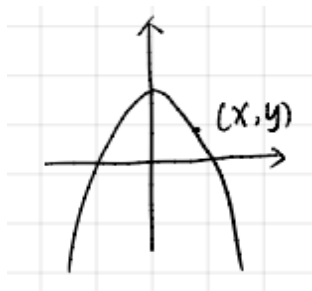
$$\min f(x) = \frac{df(x)}{dx} \Big|_{x=x_0} = 0$$

unconstrained optimization

$$f'(x) = 0 \Rightarrow x_0 \text{ which minimize } f(x)$$

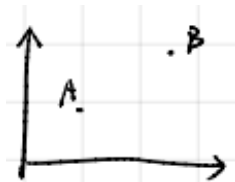
It is easy to manage, so we need to convert the constrained optimization into a non-constrained optimization

Example A constrained optimization:



$$f(x) = -2x^2 + 1, \text{ we need to find } \min d = (x^2 + y^2)^{0.5} \text{ s.t. } y = -2x^2 + 1$$

Example Suppose a curve $y = f(x)$



Find

$$\min \mathcal{L} = \int_a^b \sqrt{1 + (f'(x))^2} dx \quad \text{s.t.} \quad f(a) = y_0 \quad f(b) = y_1$$

Example find the shortest distance from the origin to the plane $x - 2y - 2z = 3$

$$\min d = x^2 + y^2 + z^2 \quad \text{s.t.} \quad x - 2y - 2z - 3 = 0$$

We can use Lagrange Multiplier. First, we convert the constrained optimization into a non-constrained optimization.

$$J = (x^2 + y^2 + z^2) + \lambda(x - 2y - 2z - 3)$$

In which, λ is Lagrange Multiplier

$$\begin{aligned} \min J &\Rightarrow (x^*, y^*, z^*) \\ (x, y, z, \lambda) \end{aligned}$$

Then we can get

$$\frac{\partial J}{\partial x} = 0 \quad \frac{\partial J}{\partial y} = 0 \quad \frac{\partial J}{\partial z} = 0 \quad \frac{\partial J}{\partial \lambda} = 0$$

Then calculate these equations.

$$\begin{aligned} 2x + \lambda &= 0 & 2z - 2\lambda &= 0 \\ 2y - 2\lambda &= 0 & x - 2y - 2z - 3 &= 0 \end{aligned}$$

After that,

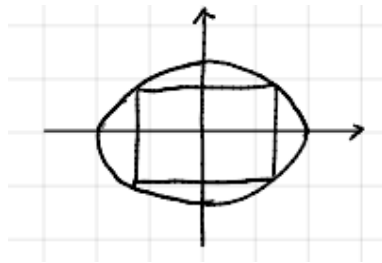
$$\Rightarrow x = \frac{\lambda}{2} \quad y = \lambda \quad z = \lambda \quad \text{代入} \quad x - 2y - 2z - 1 = 0$$

$$\Rightarrow \frac{\lambda}{2} + 2\lambda + 2\lambda + 3 = 0$$

$$\Rightarrow \lambda = -\frac{2}{3}$$

So minimum point is $\left(\frac{1}{3}, -\frac{2}{3}, -\frac{2}{3}\right)$

Example There is a ellipse, find the largest rectangle inside.



$$\min \int_{\square} = \min(4xy) \quad \text{s.t.} \quad \frac{x^2}{a^2} + \frac{y^2}{b^2} \geq 1 \Rightarrow \frac{x^2}{a^2} + \frac{y^2}{b^2} - 1 \geq 0$$

Then we can get

$$J = 4xy + \lambda \left(\frac{x^2}{a^2} + \frac{y^2}{b^2} - 1 \right)$$

Then,

$$\begin{aligned} \frac{\partial J}{\partial x} &= 4y + \frac{\lambda}{a^2} 2x = 0 \Rightarrow 2y + \frac{\lambda}{a^2} x = 0 \\ \frac{\partial J}{\partial y} &= 4x + \frac{\lambda}{b^2} \cdot 2y = 0 \Rightarrow 2x + \frac{\lambda}{b^2} y = 0 \\ \frac{\partial J}{\partial \lambda} &= \frac{x^2}{a^2} + \frac{y^2}{b^2} - 1 = 0 \Rightarrow \frac{x^2}{a^2} + \frac{y^2}{b^2} = 1 \end{aligned}$$

3. Change of variables

Example Wave equation $\frac{\partial^2 F}{\partial x^2} - \frac{1}{v^2} \frac{\partial^2 F}{\partial t^2} = 0$ and $F(x, t)$

Let

$$\begin{aligned} r &= x + vt = r(x, t) \\ s &= x - vt = s(x, t) \end{aligned}$$

Then,

$$x = \frac{r + s}{2} \quad t = \frac{r - s}{2v}$$

which is a coordinate transform. After that,

$$F(x, t) \rightarrow F(x(r, s), t(r, s)) \rightarrow F(r(x, t), s(x, t))$$

$$\begin{aligned} \frac{\partial F}{\partial x} &= \frac{\partial F}{\partial r} \frac{\partial r}{\partial x} + \frac{\partial F}{\partial s} \frac{\partial s}{\partial x} \\ \frac{\partial F}{\partial x} &= \frac{\partial F}{\partial r} + \frac{\partial F}{\partial s} \end{aligned}$$

And,

$$\begin{aligned} \frac{\partial F}{\partial t} &= \frac{\partial F}{\partial r} \cdot \frac{\partial r}{\partial t} + \frac{\partial F}{\partial s} \cdot \frac{\partial s}{\partial t} \\ \frac{\partial F}{\partial t} &= \frac{\partial F}{\partial r} v + \frac{\partial F}{\partial s} (-v) \end{aligned}$$

Suppose $g = \frac{\partial F}{\partial r}$ $h = \frac{\partial F}{\partial s}$, we can get,

$$\begin{aligned}
\frac{\partial}{\partial x} \left(\frac{\partial F}{\partial x} \right) &= \frac{\partial g}{\partial r} \cdot \frac{\partial r}{\partial x} + \frac{\partial g}{\partial s} \cdot \frac{\partial s}{\partial x} + \frac{\partial h}{\partial r} \cdot \frac{\partial r}{\partial x} + \frac{\partial h}{\partial s} \cdot \frac{\partial s}{\partial x} \\
&= \frac{\partial g}{\partial r} + \frac{\partial g}{\partial s} + \frac{\partial h}{\partial r} + \frac{\partial h}{\partial s} \\
&= \frac{\partial^2 F}{\partial r^2} + \frac{\partial^2 F}{\partial r \partial s} + \frac{\partial^2 F}{\partial r \partial s} + \frac{\partial^2 F}{\partial s^2}
\end{aligned}$$

And,

$$\frac{\partial}{\partial t} \left(\frac{\partial F}{\partial t} \right) = v^2 \left(\frac{\partial^2 F}{\partial r^2} - 2 \frac{\partial^2 F}{\partial r \partial s} + \frac{\partial^2 F}{\partial s^2} \right)$$

With $\frac{\partial^2 F}{\partial x^2} - \frac{1}{v^2} \frac{\partial^2 F}{\partial t^2} = 0$, we can get $\frac{\partial^2 F}{\partial r \partial s} = 0$

$$\frac{\partial}{\partial s} \left(\frac{\partial F}{\partial r} \right) = 0 \Rightarrow \frac{\partial F}{\partial r} = C(s) + C$$

Then $F = f(s) + g(r)$

Finally we can get

$$F(x, t) = f(x - vt) + g(x + vt)$$

Example Laplace equation $\frac{\partial^2 F}{\partial x^2} + \frac{\partial^2 F}{\partial y^2} = 0$

Let

$$\begin{aligned}
x &= r \cos \theta \\
y &= r \sin \theta
\end{aligned}$$

Then,

$$\begin{aligned}
\frac{\partial F}{\partial r} &= \frac{\partial F}{\partial x} \frac{\partial x}{\partial r} + \frac{\partial F}{\partial y} \frac{\partial y}{\partial r} = \frac{\partial F}{\partial x} \cos \theta + \frac{\partial F}{\partial y} \sin \theta \\
\frac{\partial F}{\partial \theta} &= \frac{\partial F}{\partial x} \cdot \frac{\partial x}{\partial \theta} + \frac{\partial F}{\partial y} \cdot \frac{\partial y}{\partial \theta} = \frac{\partial F}{\partial x} \cdot r \cdot (-\sin \theta) + \frac{\partial F}{\partial y} \cdot r \cos \theta
\end{aligned}$$

See page 230 of the English reference book for steps to follow

The final result is

$$\frac{1}{r} \frac{\partial}{\partial r} \left(r \frac{\partial F}{\partial r} \right) + \frac{1}{r^2} \frac{\partial^2 F}{\partial \theta^2} = 0$$